

Program : Diploma in Artificial Intelligence & Machine Learning	
Course Code : 3349	Course Title: Logic System Design Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Students will be able to demonstrate various digital circuits using trainer kit and ICs and create various systems by using digital logic

Course Prerequisites:

Topic	Course name	Semester
Basics of Electrical and Electronics Engineering	Fundamentals of Electrical and Electronics Engineering	2

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Develop basic logic functions using TTL logic gates	12	Applying
CO2	Construct Combinational logic circuits using logic gates	15	Applying
CO3	Construct & test the operations of various sequential circuits	9	Applying
CO4	Open-ended experiments	6	Applying
	Lab Exam	3	

CO – PO Mapping

CO _n	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3	3	3	3	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of the Experiment	Duration (Hours)	Cognitive Level
CO1	Develop basic logic functions using TTL logic gates		
M1.01	Test the functionality & verify truth table of basic gates (IC 7404, 7408, 7432)	3	Applying
M1.02	Test the functionality & verify truth table of NAND and NOR gates (IC 7400, 7402)		Applying
M1.03	Test the functionality & verify the truth tables of logic gates XOR,XNOR.(IC 7486, 74266)	3	Applying
M1.04	Implement basic gates using NAND and NOR gates only.	3	Applying
M1.05	Build the logic circuit on breadboard to check the De-Morgan's theorem	3	Applying
CO2	Construct Combinational logic circuits using logic gates		
M2.01	Construct Half Adder and Half Subtractor using basic gates	3	Applying
M2.02	Construct Half Adder and Half Subtractor using universal gates	3	Applying
M2.03	Construct Binary to Gray and Gray to Binary code converters	3	Applying
M2.04	Construct 4 to 1 Multiplexer and 1 to 4 DeMultiplexer using Logic gates.	3	Applying
M2.05	Demonstrate the use of MUX IC 74153, Decoder/Demux IC 74139.	3	Applying
	Lab Exam -1	1.5	

CO3	Construct & test the operations of various sequential circuits		
M3.01	Construct, Test and investigate the operation of D Flip-Flop & T FlipFlop IC's	3	Applying
M3.02	Construct, Test and investigate the operation of SR Flip-Flop & JK Flip-Flop IC's	3	Applying
M3.03	Use IC7490 to construct decade counter	3	Applying
CO4	Open Ended Experiments**	6	Applying
	Lab Exam – II	1.5	

**** - Suggested Open Ended Experiments**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

1. Implement full adder
2. Implement full subtractor
3. Implement any logic circuit

Text / Reference

T/R	Book Title/Author
T1	Digital principles & Applications Albert Paul Malvino& Donald P. LeachMcGraw Hill Education; Eighth edition ISBN: 978-9339203405
T2	Digital Electronics, Roger L. TokheimMacmillian, McGraw-Hill Education (ISE Editions);International 2 Revised edition ISBN: 978-0071167963 Macmillian
T3	Digital Electronics–an introduction to theory and practice, William H. Gothmann, Prentice Hall India Learning PrivateLimited; 2 edition ISBN: 978- 8120303485
R1	Fundamentals of LogicDesign, Charles H. Roth Jr. Jaico Publishing House; First edition ISBN: 978-8172247744
R2	Digital Electronics, R. Anand, Khanna Publications, New Delhi (Edition 2018) ISBN: 978-93-82609445

Online Resources

Sl.No	Website Link
1	https://www.iitg.ac.in/cseweb/vlab/Digital-System-Lab/experiments.php
2	https://en.wikibooks.org/wiki/Digital_Electronics/
3	https://www.geeksforgeeks.org/digital-electronics-logic-design-tutorials/
4	https://www.allaboutcircuits.com/textbook/digital/
5	https://www.researchgate.net/publication/264005171_Digital_Electronics